

MATTHEW NATANAEL

Bandung, West Java, Indonesia

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Education

BINUS University

Bandung, Indonesia

Bachelor of Computer Science (Undergraduate)

Sep. 2024 – Present

- **Relevant Coursework:** Cloud Services & Computing, Database Technology, Web Based Programming, Data Structures & Algorithms, Object-Oriented Programming, Mobile Programming, Artificial Intelligence.

Technical Skills

Languages: Python, SQL, PHP, JavaScript, C, HTML/CSS

Frameworks & Libraries: Laravel, Node.js, Discord.js, Nuxt.js, Vue.js, Tailwind CSS, Bootstrap

Developer Tools & Infrastructure: Git, GitHub, MySQL, VS Code, Vercel

Research & Publications

Adaptive Neuro-Fuzzy Based Forwarding Strategy for High-Speed VNDN | *ndnSIM, NS-3, Python*

- Co-authored research proposing a Hybrid-AI (Neuro-Fuzzy) forwarding strategy to optimize handover and reduce latency in Vehicular Named Data Networking (VNDN) for 6G environments.
- Analyzed simulation dataset and evaluated Quality of Service (QoS) metrics—achieving ultra-low latency (10.44 ms) and optimized throughput under extreme vehicle mobility.
- Drafted the academic manuscript, structured mathematical/fuzzy rule models, and interpreted network performance metrics using Python.
- *Status: Submitted / Under Review.*

Featured Projects

Deepfake Detection System | *Python, YOLOv8*

- Conducted a comparative analysis of object detection algorithms to identify manipulated media and deepfakes.
- Trained and deployed a computer vision model utilizing YOLOv8, hosted as a web application on Hugging Face Spaces.

Takeibo Financial Bot | *Node.js, JavaScript, MySQL, Discord.js, Logic Automation*

- Engineered an automated bot to log, categorize, and aggregate daily transaction data based on Japanese budgeting methodology.
- Designed relational database schemas and optimized SQL queries to support reliable data persistence and transaction auditing.

E-Archive Management System | *Fullstack Web, PHP, Laravel, MySQL, Bootstrap*

- Collaborated in a team to build a centralized web-based digital archiving platform to digitize and manage physical institutional records.
- Designed relational MySQL database schemas and optimized queries to ensure high-performance CRUD workflows and rapid document retrieval.
- Implemented Role-Based Access Control (RBAC) using Laravel to enforce strict data privacy and govern authorization levels for sensitive archives.

Organizational Experience

Bina Nusantara Computer Club (BNCC)

Bandung, Indonesia

Manager of External Event Organizer

Oct. 2025 – Present

- Spearheaded event division operations and technical logistics for **BNCC TechDive 2025: Cloud-Powered Transformation**.
- Led committee coordination across rundown management, vendor procurement, and live operational execution.

Bina Nusantara Computer Club (BNCC)

Bandung, Indonesia

Activist of External Event Organizer

Jan. 2025 – Sep. 2025

- Served as event committee member responsible for conceptualizing event design, drafting detailed rundowns, and listing potential guest speakers for **Smart Learning Based on AI: Roadmap to Tomorrow**.
- Operated broadcasting tools and technical presentation infrastructure, ensuring zero latency and smooth execution during sessions.